

k-epsilon Turbulence Modelling

Two equation turbulence model

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Abstract — Turbulent modelling has been in active research for over multiple decades now. It is more interesting and more applicable in everyday life compared to laminar flows. Their main properties include unsteadiness, irregular chaotic motion, non-linearity and vorticity [1]. There are several turbulent models in use which are suitable for different cases. The k-epsilon model is one such popular turbulence model used in computational fluid dynamics to simulate turbulent flow. It is based on the Reynolds-averaged Navier-Stokes equations and models the turbulent kinetic energy and dissipation rate as transport equations. It is based on the concept of "energy-cascade", where large eddies break down into smaller eddies and then they release energy by viscous dissipation at the end [2]. In this report, the basic principles of the k-epsilon model, our implementation along with the associated findings will be discussed. Merge request for the submission can be found here.

1 Introduction

In this section, we provide a brief overview of the k-epsilon ($k-\epsilon$) model and its origin, as well as the necessary equations for the model. The Smagorinsky approach had a major disadvantage that the somewhat satisfactory results being obtained are only for a very low mesh size [3]. To overcome this drawback, a $k-\epsilon$ model was suggested by Launder Spalding, 1974 [4]. The $k-\epsilon$ model is a two-equation model, which means that two additional variables k and ϵ are introduced to model the Reynolds stresses. Of these, k denotes the turbulent kinetic energy and ϵ the dissipation rate. The turbulent eddy viscosity now depends on these two variables,

$$\nu_t = f(k, \epsilon) \quad (1)$$

It can be defined as,

$$\nu_t = c_\mu \frac{k^2}{\epsilon} \quad (2)$$

where, c_μ is a constant

The following are the transport equations for k and ϵ respectively,

$$\frac{\partial k}{\partial t} + \vec{U} \text{grad} k - \frac{\nu_t}{2} [\text{grad} \vec{U} + \text{grad} \vec{U}^T]^2 - \text{div}(\nu_t \text{grad} k) + \epsilon = 0 \quad (3)$$

$$\frac{\partial \epsilon}{\partial t} + \vec{U} \text{grad} \epsilon - \frac{c_1}{2} k [\text{grad} \vec{U} + \text{grad} \vec{U}^T]^2 - \text{div}(\frac{c_\epsilon}{c_\mu} \nu_t \text{grad} \epsilon) + c_2 \frac{\epsilon^2}{k} = 0 \quad (4)$$

The momentum equations now in component form are as follows,

$$\frac{\partial U}{\partial t} + \frac{\partial U^2}{\partial x} + \frac{\partial(UV)}{\partial y} + \frac{\partial P}{\partial x} + \frac{2}{3} \frac{\partial k}{\partial x} - \frac{\partial}{\partial x} (2\nu^* \frac{\partial U}{\partial x}) - \frac{\partial}{\partial y} (\nu^* (\frac{\partial U}{\partial y} + \frac{\partial V}{\partial x})) = G_x \quad (5)$$

$$\frac{\partial V}{\partial t} + \frac{\partial UV}{\partial x} + \frac{\partial V^2}{\partial y} + \frac{\partial P}{\partial y} + \frac{2}{3} \frac{\partial k}{\partial y} - \frac{\partial}{\partial x} (\nu^* (\frac{\partial U}{\partial y} + \frac{\partial V}{\partial x})) - \frac{\partial}{\partial y} (2\nu^* \frac{\partial V}{\partial y}) = G_y \quad (6)$$

2 Theory

In this section, we provide a detailed explanation of the mathematical principles of the k- ε model, including the governing equations and the assumptions made. We still operate in the in-compressible domain and the turbulence is assumed to be isotropic, meaning they are uniform in all directions.

The next important aspect is boundary condition and we need to define them as the k- ε model is not valid near the solid walls. The reason for this is that in the vicinity of the walls, the local Reynolds number is low due to small velocities near the wall and the k- ε model is only valid for higher Reynolds numbers.

In turbulent flows, a thin boundary layer is formed near the walls and it comprises of three regions:

- Laminar zone: Region immediately close to the wall. Flow velocity is very small due to the no-slip condition on the wall
- Transition zone: After the laminar layer. Here the flow becomes locally unstable and the velocity profile is logarithmic
- Turbulent zone: In this zone, the viscosity is insignificant

There are two techniques to determine the flow behavior in the boundary:

1. Wall function

In this method, the values are determined from approximate calculations of the flow profile in the wall region and also empirically (experimental data). The flow is discretized and computed only in the turbulent region and the wall function operates on the laminar and the logarithmic layer. We impose certain boundary conditions on the boundary of the turbulent region, which lies at a distance δ from the nearest wall of the actual flow domain. The wall function is implicitly given by:

$$\Phi_t \left| \nu \frac{\Phi_t}{n} \right|^{-1/2} = 2.44 \log \left(\frac{\delta}{\sqrt{\nu}} \left| \frac{\Phi_t}{n} \right|^{1/2} \right) + 5.5 \quad (7)$$

2. Low-Reynolds-number model

In this technique, we adjust the coefficients of the original k- ε model in such a way that damping effects near the wall or boundary layer are accounted for. We use certain functions depending on the local Reynolds number in addition to the existing constants to account for the damping in the boundary layer or near the wall. A problem with this model is that good resolution of mesh near the wall is required.

3 Implementation

Having looked at both the boundary condition techniques for the k- ε model, we now have to decide on a particular method. The wall-function technique contains an implicit equation which was hard to implement. There also arises problem for separating flows. Also, it is quite unclear as to how regions of reversed flow are to be treated. It was quite evident to us that the low-Re model was more practical to implement. Further details on the model and parameter values are discussed in section 3.2

3.1 Stencils and additional classes

We discuss the implementation of the k- ε model with respect to our existing code framework. We inherit a new child of the **Simulation()** class, termed, **TurbulentSimulationKE()** and extend it further.

Also, for our model, four key parameters are used: turbulent kinetic energy k , dissipation rate ε , minimum distance from wall h and viscosity ν_T . However, h remains unchanged from our previous turbulence implementation. So, in addition to the inherited stencils, we apply new Stencils: **nuTStencilKE()**, **kStencil()**, **epsilonStencil()** and a **dtStencilKE()**, to accommodate our new model with the existing **hStencil()** being templated.

Additionally, we also need **MaxKStencil()** and **MaxEpsStencil()**, that respectively return the maximum value of k and ε from their stencils. Consequently, all Stencils have their respective iterators.

In **FGHStencil()**, we override the apply method to hold the TurbulentFlowFieldKE. A new flowfield is also created to hold the values of k and ε , hence TurbulentFlowField header file. A new **TurbulentVTKStencilKE()** was also created to accommodate the two new variables.

3.2 Initialization, Boundary and Initial Conditions

Apart from the normal initialization, the flow parameters k and ε are initialized in separate Stencils, **Initk-FlowFieldStencil()** and **InitEpsilonFlowFieldStencil()** respectively. It sets the value of k to 0.00015 and ε to 1.35e-5 throughout the domain.

We now discuss about the boundary conditions for the implemented low-Re boundary model. The constants along with other parameters need to be provided through the input file. The values of which are chosen from literature as follows[5]:

$$c_1 : 0.126 \quad c_2 : 1.92 \quad c_\mu : 0.09 \quad c_\varepsilon : 0.07 \quad (8)$$

We discretize simple Dirichlet and Neumann boundary conditions for k and ε . However owing to the small velocities near the wall, the local Reynolds number is low. Therefore, we adjust the coefficients so that the damping effects are accounted for. The model constants are then multiplied with some functions that in turn depends on these two local Reynolds numbers given as[6]:

$$R_t = \frac{k^2}{\nu \varepsilon} \quad R_d = \sqrt{k} \frac{d}{\nu} \quad (9)$$

$$f_\mu = (1 - \exp(-0.0165 R_d))^2 (1 + \frac{20.5}{R_t}) \quad (10)$$

$$f_1 = 1 + (\frac{0.005}{f_\mu})^3 \quad f_2 = 1 - \exp(-R_t^2) \quad (11)$$

For boundary conditions for $\vec{U}$, k and ε , along solid walls we use,

$$\vec{U} = 0 \quad k = 0 \quad \frac{\partial \varepsilon}{\partial n} = 0 \quad (12)$$

and at Outflow we use,

$$\frac{\partial \vec{U}}{\partial n} = 0 \quad \frac{\partial k}{\partial n} = 0 \quad \frac{\partial \varepsilon}{\partial n} = 0 \quad (13)$$

with suitable inflow functions based on our test case.

3.3 Discretization

The flux equations are now discretized as follows,

$$\hat{F}_{i,j} = U_{i,j} + dt \left(2 \left[\frac{\partial}{\partial x} \left(\nu^* \frac{\partial U}{\partial x} \right) \right]_{i,j} + \left[\frac{\partial}{\partial y} \left(\nu^* \left(\frac{\partial U}{\partial y} + \frac{\partial V}{\partial x} \right) \right) \right]_{i,j} - \frac{2}{3} \left[\frac{\partial k}{\partial x} \right]_{i,j} - \left[\frac{\partial (U^2)}{\partial x} \right]_{i,j} - \left[\frac{\partial (UV)}{\partial y} \right]_{i,j} + g_x \right) \quad (14)$$

$$\hat{G}_{i,j} = V_{i,j} + dt \left(2 \left[\frac{\partial}{\partial y} \left(\nu^* \frac{\partial V}{\partial y} \right) \right]_{i,j} + \left[\frac{\partial}{\partial x} \left(\nu^* \left(\frac{\partial U}{\partial y} + \frac{\partial V}{\partial x} \right) \right) \right]_{i,j} - \frac{2}{3} \left[\frac{\partial k}{\partial y} \right]_{i,j} - \left[\frac{\partial (V^2)}{\partial y} \right]_{i,j} - \left[\frac{\partial (UV)}{\partial x} \right]_{i,j} + g_y \right) \quad (15)$$

The discretization of the normal convective and diffusive terms are exactly the same as in the previous turbulence model. The only new term is the discretization of k , which is a forward difference scheme.

The discretization of k and ε at the cell centers are as follows:

$$k_{i,j}^{(n+1)} = k_{i,j}^{(n)} + \delta t \left(\left[\frac{\partial}{\partial x} \left(\nu_T \frac{\partial k}{\partial x} \right) \right]_{i,j} + \left[\frac{\partial}{\partial y} \left(\nu_T \frac{\partial k}{\partial y} \right) \right]_{i,j} - \left[\frac{\partial(Uk)}{\partial x} \right]_{i,j} - \left[\frac{\partial(Vk)}{\partial y} \right]_{i,j} + \frac{1}{2} [\nu_T]_{i,j} [|\text{grad} \vec{U} + \text{grad} \vec{U}^T|^2]_{i,j} - \varepsilon_{i,j} \right) \quad (16)$$

$$\varepsilon_{i,j}^{(n+1)} = \varepsilon_{i,j}^{(n)} + \delta t \left(\frac{c_\varepsilon}{c_\mu} \left[\frac{\partial}{\partial x} \left([f_\mu]_{i,j} \nu_T \frac{\partial \varepsilon}{\partial x} \right) \right]_{i,j} + \frac{c_\varepsilon}{c_\mu} \left[\frac{\partial}{\partial y} \left([f_\mu]_{i,j} \nu_T \frac{\partial \varepsilon}{\partial y} \right) \right]_{i,j} - \left[\frac{\partial(U\varepsilon)}{\partial x} \right]_{i,j} - \left[\frac{\partial(V\varepsilon)}{\partial y} \right]_{i,j} + \frac{c_1 [f_1]_{i,j}}{2} k_{i,j} [|\text{grad} \vec{U} + \text{grad} \vec{U}^T|^2]_{i,j} - \frac{c_2 [f_2]_{i,j}}{2} \frac{\varepsilon_{i,j}^2}{k_{i,j}} \right) \quad (17)$$

where,

$$[|\text{grad} \vec{U} + \text{grad} \vec{U}^T|^2]_{i,j} = 4 \left[\frac{\partial U}{\partial x} \right]_{i,j}^2 + 2 \left(\left[\frac{\partial U}{\partial y} \right]_{i,j} + \left[\frac{\partial V}{\partial x} \right]_{i,j} \right)^2 + 4 \left[\frac{\partial V}{\partial y} \right]_{i,j}^2 \quad (18)$$

3.4 Algorithm

Algorithm 1: k - ε Turbulence Modelling

$t = 0, n = 0$

Initial values to U, V, k , and ε

while $t < t_{end}$ **do**

 Set the boundary values for U, V, k , and ε ;

 Compute k^{n+1} and ε^{n+1} according to their respective transport equations ;

 Compute F^n and G^n ;

 Compute the right-hand side of the pressure equation ;

 Set $it = 0$;

while $it < it_{max}$ and residual $>$ tolerance **do**

 Solve Pressure Poisson equation ;

 Compute the residual norm of the pressure equation ;

$it = it + 1$;

end

 Compute U^{n+1} and V^{n+1} ;

$t = t + dt$;

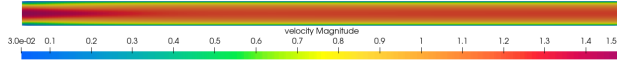
$n = n + 1$;

end

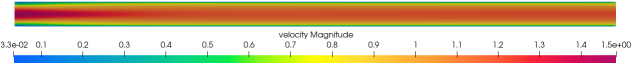
4 Results and Validation

4.1 Channel 2D

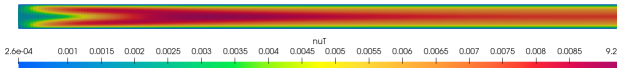
4.1.1 Results



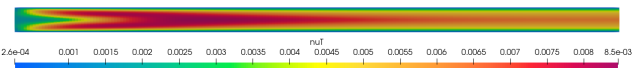
(a) Velocity Contour $Re = 5600$



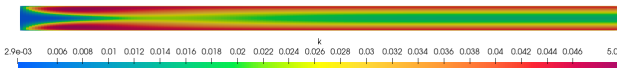
(b) Velocity Contour $Re = 13750$



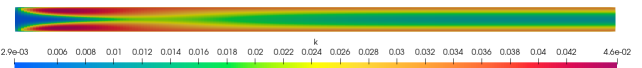
(c) ν_t Contour $Re = 5600$



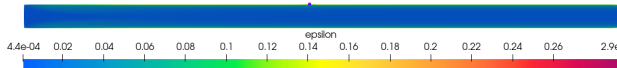
(d) ν_t Contour $Re = 13750$



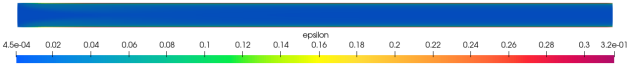
(e) k Contour $Re = 5600$



(f) k Contour $Re = 13750$



(g) ϵ Contour $Re = 5600$



(h) ϵ Contour $Re = 13750$

The results for two different Reynolds number 5600 and 13750. The value of turbulent kinetic energy k is observed to be high near to boundary, zero at the boundary and it decreases towards the center of the channel. The turbulence dissipation rate ϵ is observed to be maximum at the boundary and close to zero in the rest of the domain.

4.1.2 Validation for Channel 2D

Here some validation results are shown

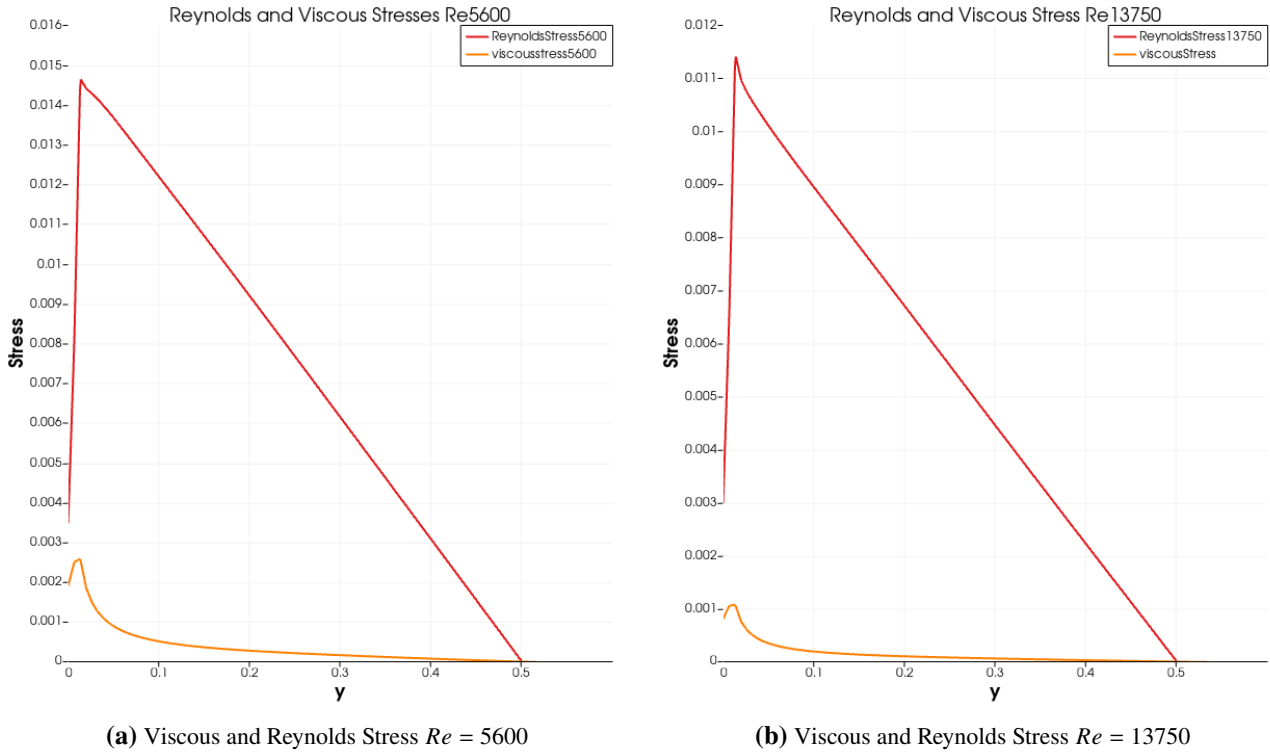


Figure 2 Shear Stress Plots

The above two figures show the Reynolds stress and Viscous stress profiles for two different Reynolds numbers. It can be seen that as the Reynolds number is increased, the viscous stress lowers, which is as expected. However, the Reynolds stress also appear to decrease, this may be in part due to less mesh resolution near the wall.

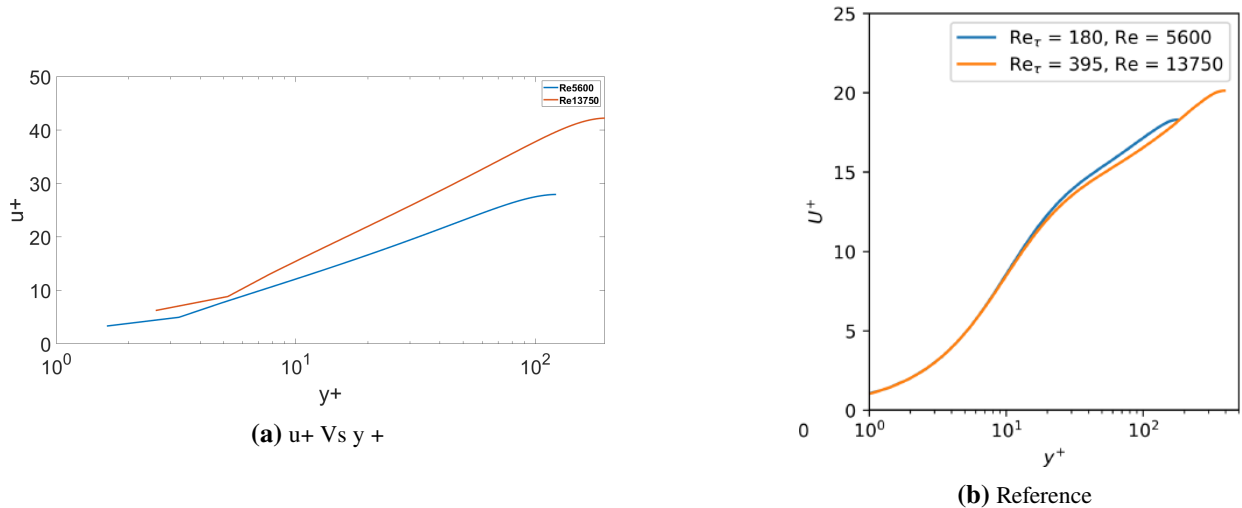


Figure 3 u^+, y^+ Plots comparison

This is a comparison between the reference $u^+ Vs y^+$ and our implementation. The trend of the graph in Fig. 3a matches to the trend in Fig. 3b. The order of magnitudes also seem to match.

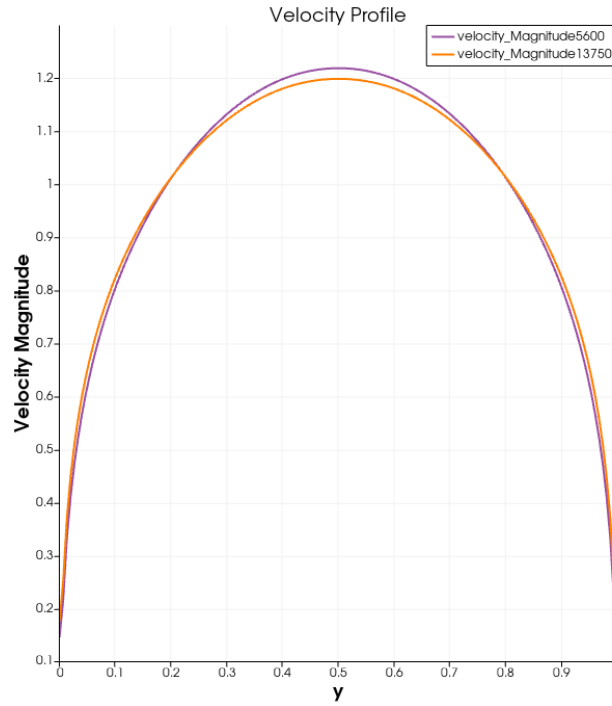
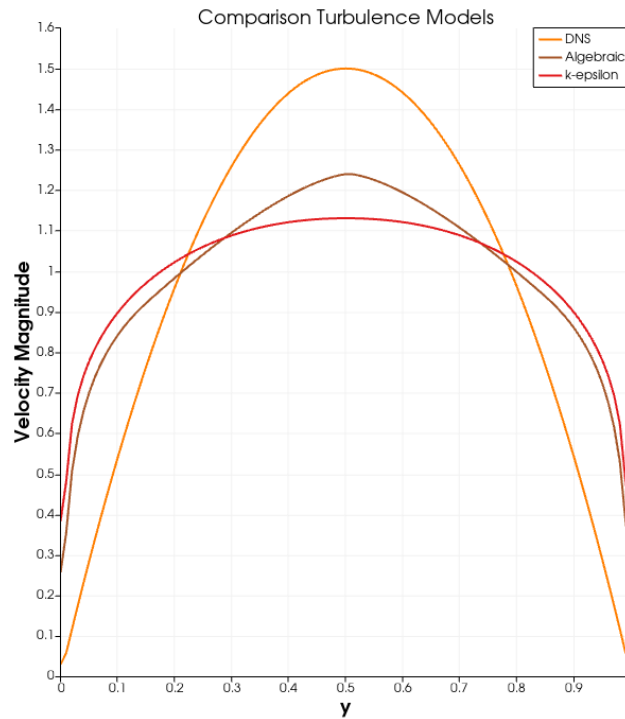


Figure 4 Velocity Plot Over line $Re = 5600, 13750$

This plot shows the comparison of velocity profiles for two Reynolds number. It can be seen that the magnitude of maximum velocity decreases as the Reynolds number is increased. This is due to increase in turbulence as Re is increased.

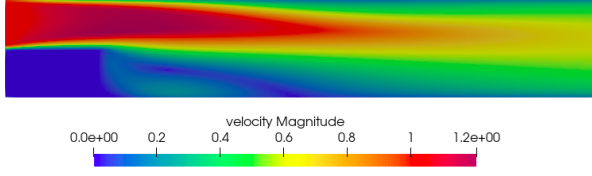


(a) Comparison DNS, Algebraic and k- ϵ turbulence model

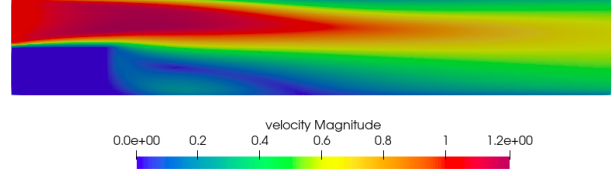
Finally, the velocity profiles of different modeling approaches are compared. The Re used for this comparison was 20000. It can be seen that the magnitude of maximum velocity decreases and the profile becomes flatter from **DNS** to **Algebraic** to **k-epsilon** turbulence model.

4.2 Backward Facing Step

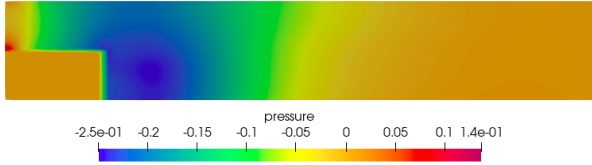
4.2.1 Results



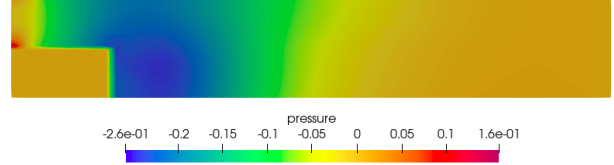
(a) Velocity Contour $Re = 30317$



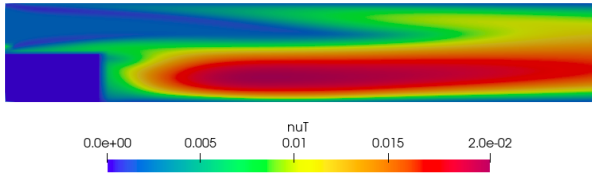
(b) Velocity Contour $Re = 47606$



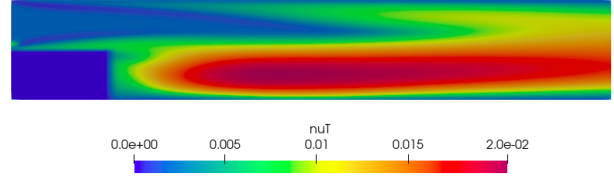
(c) Pressure Contour $Re = 30317$



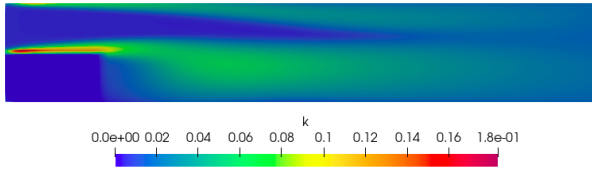
(d) Pressure Contour $Re = 47606$



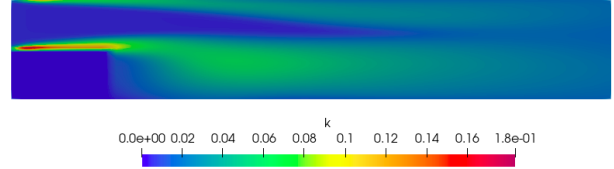
(e) nuT Contour $Re = 30317$



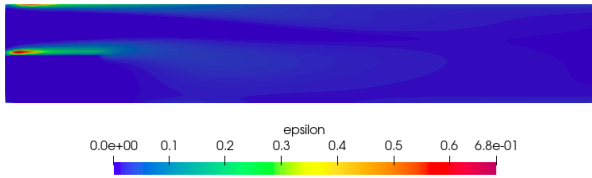
(f) nuT Contour $Re = 47606$



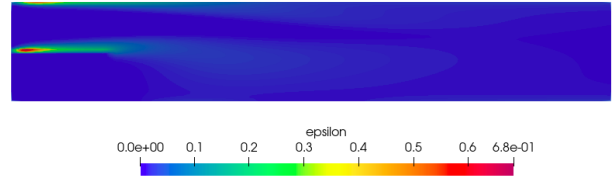
(g) k Contour $Re = 30317$



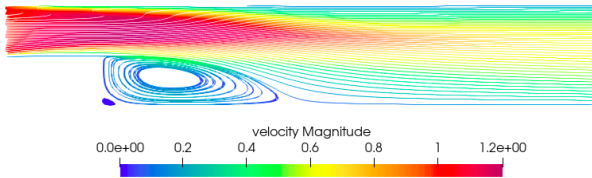
(h) k Contour $Re = 47606$



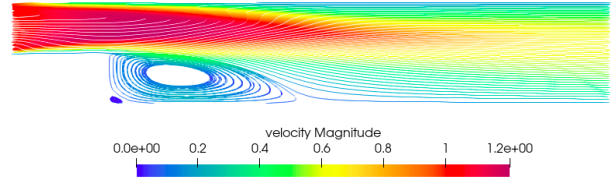
(i) Epsilon Contour $Re = 30317$



(j) Epsilon Contour $Re = 47606$



(k) Streamlines Contour $Re = 30317$



(l) Streamlines Contour $Re = 47606$

4.2.2 Validation for Backward Facing Step

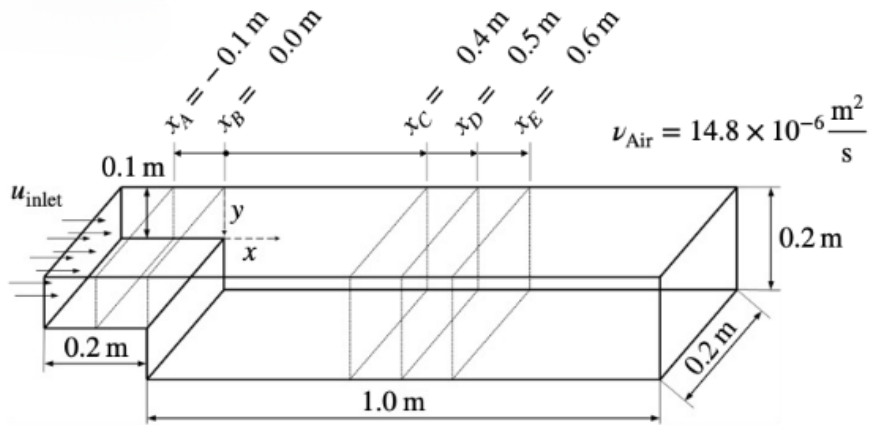
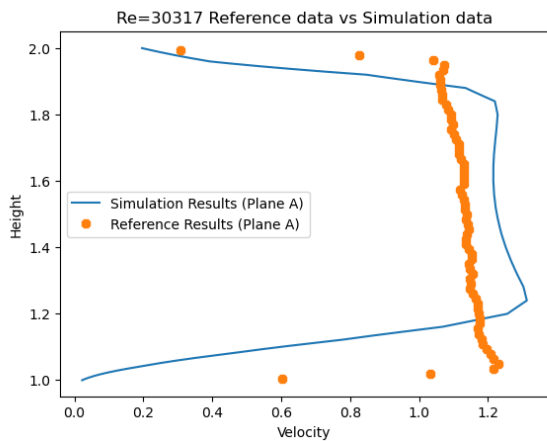
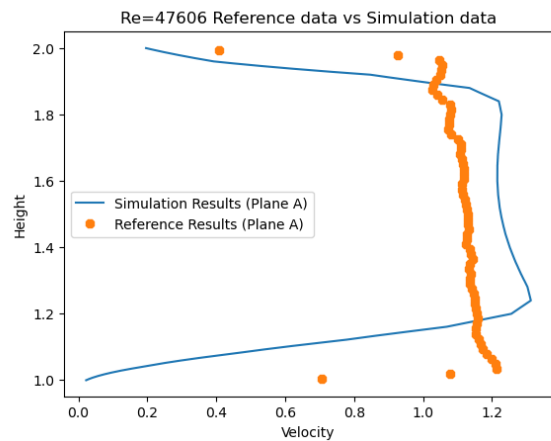


Figure 7 Backward Facing Step with all the cut-planes

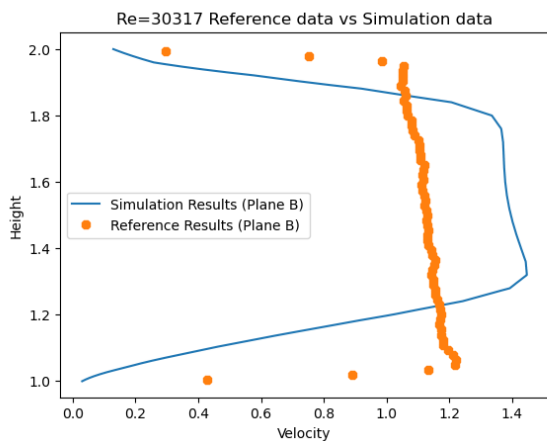
We have compared our results with the benchmark BFS data provided to us. Velocity plot over lines are compared at 5 different planes as shown in fig.7. We have used two different Reynolds Number 30317 and 47606 for this study. The results are presented in fig.8.



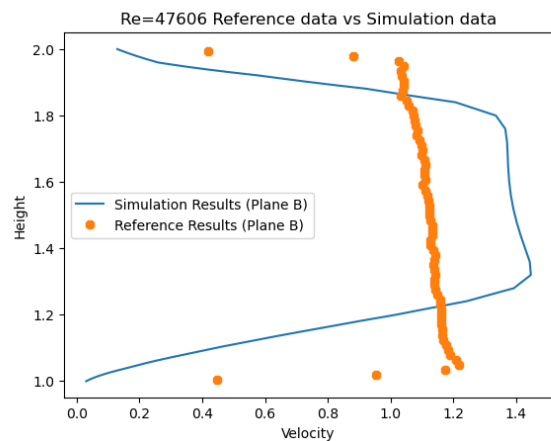
(a) Velocity plot for $Re = 30317$ at plane A



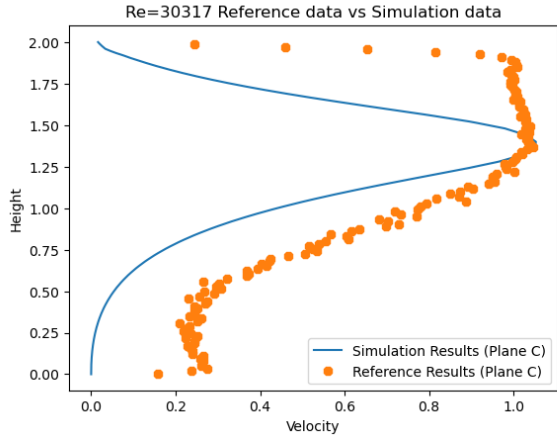
(b) Velocity plot for $Re = 47606$ at plane A



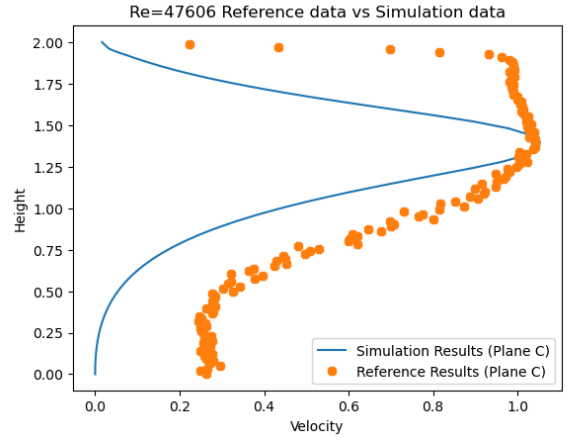
(c) Velocity plot for $Re = 30317$ at plane B



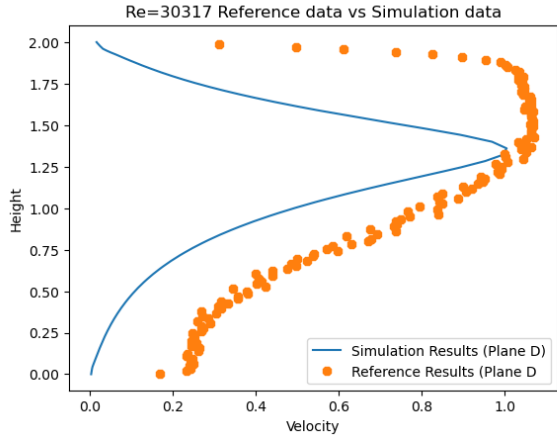
(d) Velocity plot for $Re = 47606$ at plane B



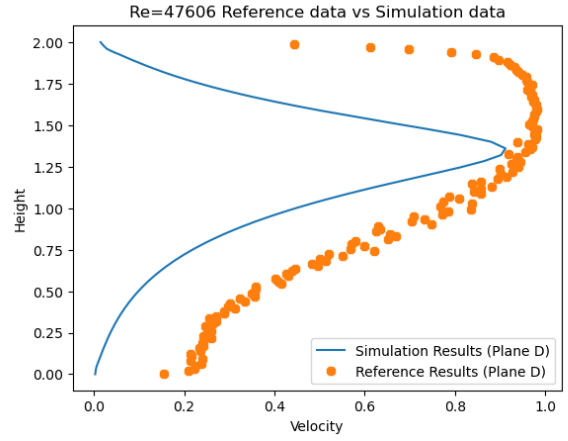
(e) Velocity plot for $Re = 30317$ at plane C



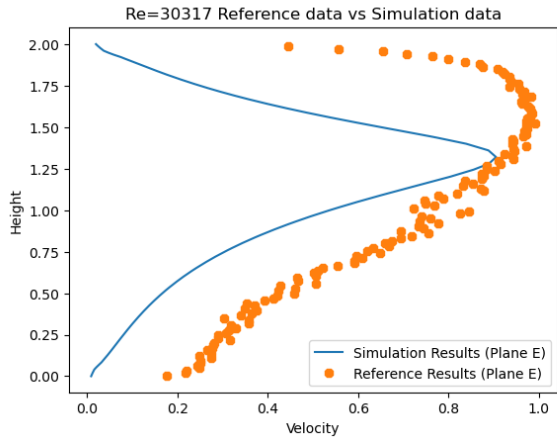
(f) Velocity plot for $Re = 47606$ at plane C



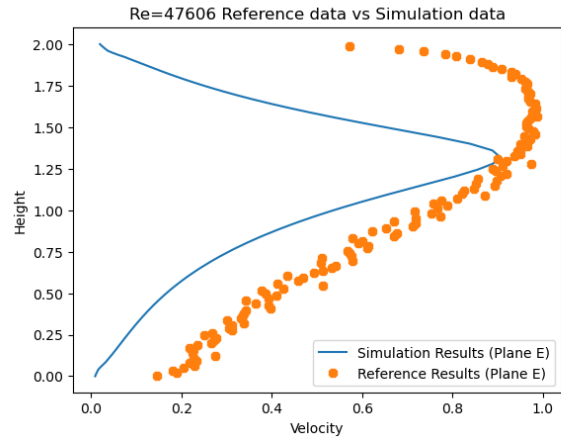
(g) Velocity plot for $Re = 30317$ at plane D



(h) Velocity plot for $Re = 47606$ at plane D



(i) Velocity plot for $Re = 30317$ at plane E



(j) Velocity plot for $Re = 47606$ at plane E

Figure 8 Results for Backward facing step at 5 different cut planes

As observed in the comparison between our results and benchmark experiment results, we have unsatisfactory match with the reference data, this could be improved with the use of proper time-step conditions, stretched mesh and an appropriate f_μ function.

5 Conclusion

We have presented our study of solving the $k-\epsilon$ model for channel flow and backward facing step. We have described the theory behind this model, the results we have obtained and also a comparison against the validation set. In our approach, we have preferred the "low-Reynolds-number-model" over the "wall-function" model. Apart from the results presented, we faced a lot of challenges in our solution approach. The main issue was because of the number of parameters involved. We were not able to validate with good accuracy as some validation report use different parameters, for example, we were unsure as to what R_τ meant (validation of channel flow, fig 3). Also, our results deviate a bit from the validation set (especially in the shear stress plots, fig 2). This can also be attributed to the lack of proper optimization of the constant values. With different boundary conditions and different constant values, it was really hard for us to get to an optimal parameter set value so as to align with the validation set. Since the used low-Re model needs a finer mesh near the boundary, usage of stretched mesh could have improved the solution. We also faced convergence problems when we use the value of f_μ as mentioned above. Its value remains unbounded, hence we had to assign a constant value of 1 to it. Another problem was the use of stretched mesh, our time-step conditions were such when used with stretched mesh gave us extremely low time-step. This was suspected due to replacing f_μ function with 1 which is not correct.

References

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